

SEQUENCE AND SERIES

Class XI Mathematics

NCERT BASED COMPLETE NOTES

Concept + Formula + Examples + NCERT Solutions

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(Mathematics)

Learn the Pattern
Understand the Formula
Solve with Confidence

Class 11 Mathematics

Chapter Overview

Sequence and Series is one of the most important chapters of Class XI Mathematics.

The basic idea is very simple:

A sequence is an ordered list of numbers.

For example,

2, 4, 6, 8, 10, ...

is a sequence.

When we add the terms of a sequence, we obtain a **series**.

For example,

$2 + 4 + 6 + 8 + 10 + \dots$

is a series.

CONCEPT

In this chapter, we will mainly study:

- Sequence
- Arithmetic Progression (A.P.)
- Geometric Progression (G.P.)
- Arithmetic Mean (A.M.)
- Geometric Mean (G.M.)
- Sum of special series

1 Sequence

A sequence is an arrangement of numbers in a definite order according to some rule.

The numbers occurring in a sequence are called its **terms**.

For example,

1, 3, 5, 7, 9, ...

is a sequence.

Here,

$$a_1 = 1, \quad a_2 = 3, \quad a_3 = 5, \quad a_4 = 7$$

where a_n denotes the n th term.

IMPORTANT FORMULA

A sequence can generally be written as

$a_1, a_2, a_3, \dots, a_n, \dots$

where a_n represents the n th term.

1.1 Finite and Infinite Sequences

A sequence having a finite number of terms is called a **finite sequence**.

Example:

2, 4, 6, 8, 10

has five terms.

A sequence having infinitely many terms is called an **infinite sequence**.

Example:

1, 2, 3, 4, 5, ...

is an infinite sequence.

1.2 Finding the General Term

The general term of a sequence is usually denoted by a_n .

Consider:

2, 4, 6, 8, 10, ...

Observe that

$$a_1 = 2(1),$$

$$a_2 = 2(2),$$

$$a_3 = 2(3),$$

$$a_4 = 2(4).$$

Therefore,

$$a_n = 2n$$

Hence the n th term is $2n$.

SMART TIP

Think like this:

If the terms are

2, 4, 6, 8, 10, ...

then each term is simply

$2 \times 1, 2 \times 2, 2 \times 3, 2 \times 4, \dots$

So,

$$a_n = 2n.$$

1.3 Example 1

Find the n th term of the sequence

3, 6, 9, 12, 15, ...

SOLUTION

We observe that

$$3 = 3(1),$$

$$6 = 3(2),$$

$$9 = 3(3),$$

$$12 = 3(4).$$

Therefore, the n th term is

$$a_n = 3n$$

1.4 Example 2

Find the n th term of

5, 10, 15, 20, ...

SOLUTION

The terms can be written as

5(1), 5(2), 5(3), 5(4), ...

Hence,

$$a_n = 5n$$

1.5 Example 3

Find the first five terms of the sequence whose general term is

$$a_n = 2n + 1.$$

SOLUTION

Put $n = 1, 2, 3, 4, 5$.

For $n = 1$,

$$a_1 = 2(1) + 1 = 3$$

For $n = 2$,

$$a_2 = 2(2) + 1 = 5$$

For $n = 3$,

$$a_3 = 2(3) + 1 = 7$$

For $n = 4$,

$$a_4 = 2(4) + 1 = 9$$

For $n = 5$,

$$a_5 = 2(5) + 1 = 11$$

Therefore,

$$3, 5, 7, 9, 11$$

CONCEPT

Very Important

If a sequence is given by a_n , then to find a particular term, simply substitute the required value of n .

For example, if

$$a_n = n^2 + 1$$

then

$$a_5 = 5^2 + 1 = 26.$$

2 NCERT-Based Practice

2.1 Question 1

Find the first five terms of the sequence whose n th term is

$$a_n = 3n - 2.$$

SOLUTION

Substitute

$$n = 1, 2, 3, 4, 5.$$

$$a_1 = 3(1) - 2 = 1,$$

$$a_2 = 3(2) - 2 = 4,$$

$$a_3 = 3(3) - 2 = 7,$$

$$a_4 = 3(4) - 2 = 10,$$

$$a_5 = 3(5) - 2 = 13.$$

Therefore,

$$\boxed{1, 4, 7, 10, 13}$$

2.2 Question 2

Find the 10th term of the sequence

$$2, 5, 8, 11, \dots$$

SOLUTION

Observe that the terms increase by 3.

The sequence can be written as

$$3(1) - 1, 3(2) - 1, 3(3) - 1, \dots$$

Hence,

$$a_n = 3n - 1.$$

For $n = 10$,

$$a_{10} = 3(10) - 1$$

$$a_{10} = 30 - 1$$

$$\boxed{a_{10} = 29}$$

3 Quick Recap**IMPORTANT FORMULA**

- A sequence is an ordered list of numbers.
- The n th term is represented by a_n .
- To find a particular term, substitute the value of n .
- A finite sequence has a limited number of terms.
- An infinite sequence continues indefinitely.

THINK AND TRY

Find the 20th term of the sequence
4, 7, 10, 13, ...

Hint: First identify the rule connecting the terms.

END OF PART 1

Next: Arithmetic Progression (A.P.)

4 Arithmetic Progression (A.P.)**CONCEPT**

A sequence in which the difference between any two consecutive terms is always the same is called an **Arithmetic Progression**, or simply an **A.P.**

4.1 Understanding A.P.

Consider the sequence

2, 5, 8, 11, 14, ...

Let us find the difference between consecutive terms:

$$5 - 2 = 3,$$

$$8 - 5 = 3,$$

$$11 - 8 = 3,$$

$$14 - 11 = 3.$$

Since the difference is always 3, this is an A.P.

IMPORTANT FORMULA

The constant difference between two consecutive terms is called the **common difference**. It is denoted by d .

$$d = a_2 - a_1$$

or

$$d = a_{n+1} - a_n$$

4.2 Standard Form of an A.P.

An A.P. is generally written as

$a, a + d, a + 2d, a + 3d, \dots$

Here,

a = first term

d = common difference

Therefore,

$$a, a + d, a + 2d, a + 3d, \dots$$

4.3 How to Identify an A.P.?

Suppose we have

7, 11, 15, 19, 23, ...

Calculate the consecutive differences:

$$11 - 7 = 4,$$

$$15 - 11 = 4,$$

$$19 - 15 = 4,$$

$$23 - 19 = 4.$$

All differences are equal.

Therefore,

The given sequence is an A.P.

with

$$d = 4$$

SMART TIP

Fast Test for A.P.

Simply subtract:

$$\text{Second term} - \text{First term}$$

If the same difference continues for every pair of consecutive terms, the sequence is an A.P.

4.4 Positive, Negative and Zero Common Difference

The value of d can be positive, negative or zero.

Case 1: $d > 0$

The terms increase.

Example:

3, 7, 11, 15, ...

Here,

$$d = 4.$$

Increasing A.P.

Case 2: $d < 0$

The terms decrease.

Example:

20, 16, 12, 8, ...

Here,

$$d = -4.$$

Decreasing A.P.

Case 3: $d = 0$

All terms are equal.

Example:

5, 5, 5, 5, ...

Here,

$d = 0$.

5 General Term of an A.P.

Consider the A.P.

$a, a + d, a + 2d, a + 3d, \dots$

The terms can be numbered as:

Term	Value
1^{st}	a
2^{nd}	$a + d$
3^{rd}	$a + 2d$
4^{th}	$a + 3d$
$\vdots$	$\vdots$
n^{th}	$a + (n - 1)d$

Therefore,

IMPORTANT FORMULA

$$a_n = a + (n - 1)d$$

This is the most important formula for finding the n th term of an A.P.

5.1 Example 1

Find the 15^{th} term of the A.P.

4, 9, 14, 19, ...

SOLUTION

First term:

$$a = 4$$

Common difference:

$$d = 9 - 4 = 5$$

We use

$$a_n = a + (n - 1)d$$

For $n = 15$,

$$a_{15} = 4 + (15 - 1)(5)$$

$$= 4 + 14(5)$$

$$= 4 + 70$$

Therefore,

$$\boxed{a_{15} = 74}$$

5.2 Example 2

Find the 20th term of the A.P.

$$7, 12, 17, 22, \dots$$

SOLUTION

Here,

$$a = 7$$

and

$$d = 12 - 7 = 5.$$

Using

$$a_n = a + (n - 1)d,$$

we get

$$a_{20} = 7 + (20 - 1)(5)$$

$$= 7 + 19(5)$$

$$= 7 + 95$$

$$\boxed{a_{20} = 102}$$

5.3 Example 3: Negative Common Difference

Find the 12th term of
30, 26, 22, 18, ...

SOLUTION

Here,

$$a = 30$$

and

$$d = 26 - 30 = -4.$$

Therefore,

$$a_{12} = a + (12 - 1)d$$

$$= 30 + 11(-4)$$

$$= 30 - 44$$

Hence,

$$\boxed{a_{12} = -14}$$

6 Finding the Number of Terms

Sometimes the first term, common difference and last term are given, but the number of terms is unknown.

Suppose

$$a_n = a + (n - 1)d$$

If the last term is l , then

$$l = a + (n - 1)d.$$

Therefore,

IMPORTANT FORMULA

$$\boxed{l = a + (n - 1)d}$$

and hence,

$$\boxed{n = \frac{l - a}{d} + 1}$$

6.1 Example 4

How many terms are there in the A.P.
5, 8, 11, ..., 80?

SOLUTION

Here,

$$a = 5, \quad d = 8 - 5 = 3, \quad l = 80.$$

Using

$$l = a + (n - 1)d,$$

we get

$$80 = 5 + (n - 1)3$$

$$75 = 3(n - 1)$$

$$25 = n - 1$$

Therefore,

$$\boxed{n = 26}$$

Hence, the A.P. contains

$$\boxed{26 \text{ terms}}.$$

7 Finding a Missing Term

Consider

5, 9, $\square$, 17, 21

Since this is an A.P., the common difference must be constant.

We have

$$d = 9 - 5 = 4.$$

Therefore, the missing term is

$$9 + 4 = 13.$$

Hence,

$$\boxed{5, 9, 13, 17, 21}$$

SMART TIP

Whenever a term is missing in an A.P., first find the common difference from the known consecutive terms.

Then use

$$\boxed{\text{Next term} = \text{Previous term} + d}$$

8 NCERT-Based Questions

8.1 Question 1

Check whether the sequence

3, 7, 11, 15, 19, ...

is an A.P. If yes, find its common difference.

SOLUTION

Calculate consecutive differences:

$$7 - 3 = 4,$$

$$11 - 7 = 4,$$

$$15 - 11 = 4,$$

$$19 - 15 = 4.$$

Since all differences are equal,

The sequence is an A.P.

and

$$d = 4.$$

8.2 Question 2

Find the 25th term of the A.P.

11, 15, 19, 23, ...

SOLUTION

Here,

$$a = 11, \quad d = 15 - 11 = 4.$$

Using

$$a_n = a + (n - 1)d,$$

$$a_{25} = 11 + (25 - 1)(4)$$

$$= 11 + 24(4)$$

$$= 11 + 96$$

Therefore,

$$a_{25} = 107$$

8.3 Question 3

Which term of the A.P.

21, 18, 15, 12, ...

is -81?

SOLUTION

Here,

$$a = 21$$

and

$$d = 18 - 21 = -3.$$

We need

$$a_n = -81.$$

Using

$$a_n = a + (n - 1)d,$$

we get

$$-81 = 21 + (n - 1)(-3)$$

$$-81 = 21 - 3(n - 1)$$

$$-102 = -3(n - 1)$$

$$34 = n - 1$$

Therefore,

$$\boxed{n = 35}$$

Hence, -81 is the

$$\boxed{35^{th} \text{ term}}.$$

8.4 Question 4

Find the first term and common difference of the A.P.

17, 13, 9, 5, ...

SOLUTION

The first term is directly visible:

$$\boxed{a = 17}$$

Now,

$$d = 13 - 17 = -4.$$

Therefore,

$$\boxed{a = 17, \quad d = -4}$$

9 Three Numbers in A.P.

Suppose three numbers are in A.P.

Instead of writing them as

$a, a + d, a + 2d,$

we can write them in a very useful form:

$$a - d, \quad a, \quad a + d$$

Why?

Because

$$a - (a - d) = d$$

and

$$(a + d) - a = d.$$

Thus, both consecutive differences are equal.

CONCEPT

If three numbers are in A.P., they can be represented as

$$a - d, \quad a, \quad a + d$$

This representation is extremely useful in algebraic problems.

9.1 Example 5

Find three numbers in A.P. whose sum is 30.

SOLUTION

Let the three numbers be

$$a - d, \quad a, \quad a + d.$$

Their sum is

$$(a - d) + a + (a + d) = 30.$$

Therefore,

$$3a = 30$$

$$a = 10.$$

Hence, the three numbers are

$$10 - d, \quad 10, \quad 10 + d$$

Notice that infinitely many triples are possible unless another condition is given.

10 Four Numbers in A.P.

Four numbers in A.P. can conveniently be written as

$$a - 3d, \quad a - d, \quad a + d, \quad a + 3d$$

Their common difference is $2d$.

For example,

2, 5, 8, 11

can be represented using

$a = 6.5$ and $d = 1.5$.

11 A.P. Formula Sheet

IMPORTANT FORMULA

First term:

$$a$$

Common difference:

$$d = a_2 - a_1$$

n th term:

$$a_n = a + (n - 1)d$$

Last term:

$$l = a + (n - 1)d$$

Number of terms:

$$n = \frac{l - a}{d} + 1$$

Three numbers in A.P.:

$$a - d, \quad a, \quad a + d$$

CHALLENGE QUESTION

The 7th term of an A.P. is 20 and the 13th term is 44. Find the first term and common difference.

Think:

Use

$$a_n = a + (n - 1)d$$

for both terms.

NEXT PART

Sum of n Terms of an A.P.

Arithmetic Mean

NCERT Exercise Problems

12 Sum of n Terms of an A.P.

Suppose we have an A.P.

$$a, a + d, a + 2d, a + 3d, \dots$$

If we add its first n terms, we get

$$S_n = a + (a + d) + (a + 2d) + \dots + [a + (n - 1)d].$$

Here, S_n represents the sum of the first n terms.

IMPORTANT FORMULA

The sum of the first n terms of an A.P. is denoted by

$$\boxed{S_n}$$

12.1 Derivation of the Formula

Let

$$S_n = a + (a + d) + (a + 2d) + \dots + (a + (n - 1)d).$$

Write the same expression in reverse order:

$$S_n = [a + (n - 1)d] + [a + (n - 2)d] + \dots + a.$$

Adding the two equations,

$$\begin{aligned} 2S_n &= [a + a + (n - 1)d] \\ &\quad + [a + d + a + (n - 2)d] \\ &\quad + [a + 2d + a + (n - 3)d] \\ &\quad + \dots \end{aligned}$$

Every pair has the same sum:

$$2a + (n - 1)d.$$

There are n such pairs.

Therefore,

$$2S_n = n[2a + (n - 1)d].$$

Hence,

IMPORTANT FORMULA

$$\boxed{S_n = \frac{n}{2}[2a + (n - 1)d]}$$

12.2 Formula Using the Last Term

We know that the n th term is

$$a_n = a + (n - 1)d.$$

If the last term is denoted by l , then

$$l = a + (n - 1)d.$$

Therefore,

$$S_n = \frac{n}{2}(a + l).$$

IMPORTANT FORMULA

If the first term is a and the last term is l , then

$$S_n = \frac{n}{2}(a + l)$$

SMART TIP

Which formula should I use?

If a , d and n are given, use

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

If a , l and n are given, use

$$S_n = \frac{n}{2}(a + l)$$

12.3 Example 1

Find the sum of the first 20 terms of the A.P.

3, 7, 11, 15, ...

SOLUTION

Here,

$$a = 3$$

$$d = 7 - 3 = 4$$

and

$$n = 20.$$

Using

$$S_n = \frac{n}{2}[2a + (n - 1)d],$$

we get

$$S_{20} = \frac{20}{2}[2(3) + (20 - 1)(4)]$$

$$= 10[6 + 19(4)]$$

$$= 10[6 + 76]$$

$$= 10(82)$$

Therefore,

$$\boxed{S_{20} = 820}$$

12.4 Example 2

Find the sum of

$$5 + 10 + 15 + \cdots + 100.$$

SOLUTION

The given sequence is
 $5, 10, 15, \dots, 100$.
Therefore,

$$a = 5, \quad d = 5, \quad l = 100.$$

First find the number of terms.
Using

$$l = a + (n - 1)d,$$

$$100 = 5 + (n - 1)5$$

$$95 = 5(n - 1)$$

$$19 = n - 1$$

Hence,

$$n = 20.$$

Now use

$$S_n = \frac{n}{2}(a + l).$$

Thus,

$$\begin{aligned} S_{20} &= \frac{20}{2}(5 + 100) \\ &= 10(105) \end{aligned}$$

$S_{20} = 1050$

12.5 Example 3

Find the sum of the first 50 natural numbers.

SOLUTION

The natural numbers form the A.P.

$1, 2, 3, 4, \dots, 50$.

Therefore,

$$a = 1, \quad d = 1, \quad n = 50.$$

Using

$$S_n = \frac{n}{2}[2a + (n - 1)d],$$

$$S_{50} = \frac{50}{2}[2(1) + (50 - 1)(1)]$$

$$= 25(2 + 49)$$

$$= 25(51)$$

Therefore,

$$\boxed{S_{50} = 1275}$$

13 Important Special Results

From the A.P. formula, we obtain some very useful results.

13.1 Sum of First n Natural Numbers

The sequence

$1, 2, 3, \dots, n$

is an A.P.

Hence,

IMPORTANT FORMULA

$$\boxed{1 + 2 + 3 + \dots + n = \frac{n(n + 1)}{2}}$$

13.2 Sum of First n Even Natural Numbers

The sequence is

$2, 4, 6, \dots, 2n$.

Here,

$$a = 2, \quad d = 2.$$

Therefore,

$$S_n = \frac{n}{2}[2(2) + (n - 1)(2)].$$

Simplifying,

$$S_n = n(n+1).$$

Thus,

IMPORTANT FORMULA

$$2 + 4 + 6 + \cdots + 2n = n(n+1)$$

13.3 Sum of First n Odd Natural Numbers

The sequence is

$$1, 3, 5, \dots, (2n-1).$$

Here,

$$a = 1, \quad d = 2.$$

Therefore,

$$S_n = \frac{n}{2}[2 + (n-1)(2)].$$

$$S_n = \frac{n}{2}(2n)$$

Hence,

IMPORTANT FORMULA

$$1 + 3 + 5 + \cdots + (2n-1) = n^2$$

CONCEPT

A beautiful result:

$$1 + 3 + 5 + \cdots + (2n-1) = n^2$$

For example,

$$1 + 3 + 5 + 7 + 9 = 25 = 5^2.$$

So the sum of the first n odd numbers is always a perfect square.

14 Arithmetic Mean

Suppose two numbers a and b are given.

Their arithmetic mean is the number obtained by taking their average.

IMPORTANT FORMULA

The arithmetic mean of two numbers a and b is

$$A = \frac{a+b}{2}$$

14.1 Example 4

Find the arithmetic mean of 12 and 20.

SOLUTION

Using

$$A = \frac{a + b}{2},$$

we get

$$A = \frac{12 + 20}{2}$$

$$A = \frac{32}{2}$$

Therefore,

$$\boxed{A = 16}$$

15 Insertion of Arithmetic Means

Suppose we want to insert m arithmetic means between two numbers a and b .

Then the complete A.P. will contain

$$m + 2$$

terms.

Therefore,

$$a_{m+2} = b.$$

Using the n th term formula,

$$b = a + (m + 1)d.$$

Hence,

$$\boxed{d = \frac{b - a}{m + 1}}$$

IMPORTANT FORMULA

If m arithmetic means are inserted between a and b , then

$$\boxed{d = \frac{b - a}{m + 1}}$$

15.1 Example 5

Insert three arithmetic means between 5 and 21.

SOLUTION

We have

$$a = 5, \quad b = 21, \quad m = 3.$$

Therefore,

$$\begin{aligned} d &= \frac{b - a}{m + 1} \\ &= \frac{21 - 5}{3 + 1} \\ &= \frac{16}{4} \\ d &= 4. \end{aligned}$$

Hence the A.P. is

$$5, 9, 13, 17, 21.$$

Therefore, the three arithmetic means are

$$\boxed{9, 13, 17}.$$

15.2 Example 6

Insert five arithmetic means between 4 and 28.

SOLUTION

Here,

$$a = 4, \quad b = 28, \quad m = 5.$$

Thus,

$$\begin{aligned} d &= \frac{28 - 4}{5 + 1} \\ &= \frac{24}{6} \\ d &= 4. \end{aligned}$$

Therefore, the A.P. is

$$4, 8, 12, 16, 20, 24, 28.$$

Hence the five arithmetic means are

$$\boxed{8, 12, 16, 20, 24}.$$

16 NCERT-Based Questions

16.1 Question 1

Find the sum of the first 30 terms of the A.P.

7, 10, 13, 16, ...

SOLUTION

Here,

$$a = 7, \quad d = 3, \quad n = 30.$$

Therefore,

$$S_{30} = \frac{30}{2}[2(7) + (30 - 1)(3)]$$

$$= 15[14 + 87]$$

$$= 15(101)$$

Hence,

$$\boxed{S_{30} = 1515}$$

16.2 Question 2

Find the sum of all odd numbers from 1 to 99.

SOLUTION

The sequence is

$1, 3, 5, \dots, 99$.

Here,

$$a = 1, \quad d = 2, \quad l = 99.$$

Find n :

$$99 = 1 + (n - 1)2$$

$$98 = 2(n - 1)$$

$$49 = n - 1$$

Therefore,

$$n = 50.$$

Now,

$$S_n = \frac{n}{2}(a + l).$$

Thus,

$$S_{50} = \frac{50}{2}(1 + 99)$$

$$= 25(100)$$

$$\boxed{S_{50} = 2500}$$

16.3 Question 3

Find the sum

$$2 + 5 + 8 + \dots + 95.$$

SOLUTION

Here,

$$a = 2, \quad d = 3, \quad l = 95.$$

Using

$$l = a + (n - 1)d,$$

we get

$$95 = 2 + (n - 1)3$$

$$93 = 3(n - 1)$$

$$31 = n - 1$$

Hence,

$$n = 32.$$

Therefore,

$$S_{32} = \frac{32}{2}(2 + 95)$$

$$= 16(97)$$

$$\boxed{S_{32} = 1552}$$

16.4 Question 4

Insert four arithmetic means between 6 and 36.

SOLUTION

Here,

$$a = 6, \quad b = 36, \quad m = 4.$$

Therefore,

$$\begin{aligned} d &= \frac{36 - 6}{4 + 1} \\ &= \frac{30}{5} \end{aligned}$$

$$d = 6.$$

Hence,

$$6, 12, 18, 24, 30, 36$$

Therefore, the four arithmetic means are

$$\boxed{12, 18, 24, 30}.$$

16.5 Question 5

The first term of an A.P. is 8 and the common difference is 3. Find the sum of the first 25 terms.

SOLUTION

Given,

$$a = 8, \quad d = 3, \quad n = 25.$$

Using

$$S_n = \frac{n}{2}[2a + (n - 1)d],$$

we get

$$\begin{aligned} S_{25} &= \frac{25}{2}[2(8) + (25 - 1)(3)] \\ &= \frac{25}{2}[16 + 72] \\ &= \frac{25}{2}(88) \\ &= 25(44) \end{aligned}$$

Therefore,

$$\boxed{S_{25} = 1100}$$

17 A.P. Master Formula Box

IMPORTANT FORMULA

First term

$$a$$

Common difference

$$d = a_2 - a_1$$

n th term

$$a_n = a + (n - 1)d$$

Last term

$$l = a + (n - 1)d$$

Number of terms

$$n = \frac{l - a}{d} + 1$$

Sum of first n terms

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

Another form

$$S_n = \frac{n}{2}(a + l)$$

Arithmetic Mean

$$A = \frac{a + b}{2}$$

m A.M.s between a and b

$$d = \frac{b - a}{m + 1}$$

18 Common Mistakes to Avoid

EXAM ALERT

1. Do not confuse a_n with S_n .
 a_n means the **n th term**,
 whereas S_n means the **sum of the first n terms**.
2. In
 $a_n = a + (n - 1)d$,
 do not write nd .
3. If the A.P. is decreasing, remember that d is negative.
4. Before using the sum formula, make sure that n is known.
5. When inserting m arithmetic means between two numbers, the total number of terms is $m + 2$.

CHALLENGE QUESTION

The sum of the first 20 terms of an A.P. is 610 and its first term is 2. Find the common difference.

Hint:

Use

$$S_n = \frac{n}{2}[2a + (n - 1)d].$$

Put

$$S_{20} = 610, \quad a = 2.$$

Then solve for d .

NEXT PART

Geometric Progression (G.P.)

General Term of G.P.

Sum of n Terms of G.P.

Geometric Mean

19 Geometric Progression (G.P.)

Till now, we studied Arithmetic Progression, where the **difference** between consecutive terms remains constant.

Now we study a different type of pattern.

Consider:

2, 6, 18, 54, 162, ...

Here,

$$\frac{6}{2} = 3, \quad \frac{18}{6} = 3, \quad \frac{54}{18} = 3.$$

The ratio between consecutive terms is constant.
This leads us to the idea of a **Geometric Progression**.

CONCEPT

A sequence of non-zero numbers is called a **Geometric Progression (G.P.)** if the ratio of any term to the preceding term is constant.
This constant ratio is called the **common ratio**.
It is denoted by r .

19.1 Common Ratio

For a G.P.

$a, ar, ar^2, ar^3, \dots$

the common ratio is

$$r = \frac{a_2}{a_1}$$

or

$$r = \frac{a_{n+1}}{a_n}$$

provided the denominator is non-zero.

19.2 Example 1

Check whether

$3, 9, 27, 81, \dots$

is a G.P.

SOLUTION

Calculate the ratios:

$$\frac{9}{3} = 3$$

$$\frac{27}{9} = 3$$

$$\frac{81}{27} = 3$$

Since all the ratios are equal,

The sequence is a G.P.

and its common ratio is

$$r = 3.$$

19.3 Example 2

Check whether

$64, 32, 16, 8, \dots$

is a G.P.

SOLUTION

We have

$$\frac{32}{64} = \frac{1}{2}$$

$$\frac{16}{32} = \frac{1}{2}$$

$$\frac{8}{16} = \frac{1}{2}.$$

Therefore, the ratio is constant.

Hence,

The sequence is a G.P.

with

$$r = \frac{1}{2}.$$

20 Standard Form of a G.P.

If the first term is a and common ratio is r , then a G.P. is

$$a, ar, ar^2, ar^3, \dots$$

Let's observe the pattern:

Term	Value
1 st	a
2 nd	ar
3 rd	ar^2
4 th	ar^3
$\vdots$	$\vdots$

Therefore, the n th term is

IMPORTANT FORMULA

$$a_n = ar^{n-1}$$

20.1 Why ar^{n-1} ?

Look carefully:

$$\begin{aligned} a_1 &= a = ar^0, \\ a_2 &= ar = ar^1, \\ a_3 &= ar^2, \\ a_4 &= ar^3. \end{aligned}$$

Therefore,

$$a_n = ar^{n-1}.$$

SMART TIP**Remember the pattern**

For A.P.:

$$a_n = a + (n - 1)d$$

For G.P.:

$$a_n = ar^{n-1}$$

A.P. → repeated **addition**G.P. → repeated **multiplication****20.2 Example 3**Find the 8th term of the G.P.

2, 6, 18, 54, ...

SOLUTION

Here,

$$a = 2$$

and

$$r = \frac{6}{2} = 3.$$

Using

$$a_n = ar^{n-1},$$

we get

$$a_8 = 2(3)^{8-1}$$

$$= 2(3^7)$$

$$= 2(2187)$$

Therefore,

$$a_8 = 4374$$

20.3 Example 4Find the 10th term of the G.P.

81, 27, 9, 3, ...

SOLUTION

First term:

$$a = 81$$

Common ratio:

$$r = \frac{27}{81} = \frac{1}{3}.$$

Therefore,

$$a_{10} = 81 \left(\frac{1}{3}\right)^9.$$

Since

$$81 = 3^4,$$

we get

$$\begin{aligned} a_{10} &= 3^4 \cdot 3^{-9} \\ &= 3^{-5} \end{aligned}$$

Hence,

$$a_{10} = \frac{1}{243}$$

21 G.P. with Negative Common Ratio

A G.P. may also have a negative common ratio.

For example,

2, -4, 8, -16, 32, ...

Here,

$$r = \frac{-4}{2} = -2.$$

The signs alternate:

+, -, +, -, +, ...

CONCEPT

If $r < 0$, consecutive terms of the G.P. have alternating signs.

For example, if

$$r = -2,$$

then

$$a, -2a, 4a, -8a, \dots$$

21.1 Case $r = 0$

If

$$r = 0,$$

then the G.P. becomes

$$a, 0, 0, 0, \dots$$

provided $a \neq 0$.

22 Finding the Number of Terms in a G.P.

Suppose the first term is a , common ratio is r and the last term is l .

Using

$$a_n = ar^{n-1},$$

we get

$$l = ar^{n-1}.$$

Therefore,

$$\frac{l}{a} = r^{n-1}.$$

If logarithms are used, then

$$\log \left(\frac{l}{a} \right) = (n-1) \log r.$$

Hence,

IMPORTANT FORMULA

$$n = 1 + \frac{\log(l/a)}{\log r}$$

when logarithms are applicable.

22.1 Example 5

How many terms are there in the G.P.

$$3, 6, 12, \dots, 384?$$

SOLUTION

Here,

$$a = 3$$

and

$$r = 2.$$

The last term is

$$l = 384.$$

Using

$$l = ar^{n-1},$$

we get

$$384 = 3(2)^{n-1}.$$

Therefore,

$$128 = 2^{n-1}.$$

Since

$$128 = 2^7,$$

we get

$$n - 1 = 7.$$

Hence,

$$\boxed{n = 8}.$$

23 Finding Missing Terms in a G.P.

Suppose

2, 6, □, 54

is a G.P.

Since

$$r = \frac{6}{2} = 3,$$

the missing term is

$$6(3) = 18.$$

Therefore,

$$\boxed{2, 6, 18, 54}$$

24 Three Numbers in G.P.

Suppose three numbers are in G.P.

A very useful representation is

$$\frac{a}{r}, \quad a, \quad ar$$

because

$$\frac{a}{a/r} = r$$

and

$$\frac{ar}{a} = r.$$

Therefore, both ratios are equal.

CONCEPT

Three positive numbers in G.P. can also be written as

$$\frac{a}{r}, \quad a, \quad ar$$

This representation is especially useful in algebraic problems.

25 Geometric Mean

Suppose two positive numbers a and b are given.

A number G is called their geometric mean if

$$a, G, b$$

are in G.P.

Therefore,

$$\frac{G}{a} = \frac{b}{G}.$$

Cross multiplying,

$$G^2 = ab.$$

Hence,

IMPORTANT FORMULA

$$G = \sqrt{ab}$$

for positive a and b .

25.1 Example 6

Find the geometric mean between 4 and 25.

SOLUTION

Using

$$G = \sqrt{ab},$$

we get

$$G = \sqrt{4 \times 25}$$

$$= \sqrt{100}$$

Therefore,

$$\boxed{G = 10}.$$

Check:

$$4, 10, 25$$

is a G.P. because

$$\frac{10}{4} \neq \frac{25}{10}.$$

Thus, 4, 10, 25 are **not** in G.P.

Hence, we must be careful!

Actually, the geometric mean is obtained from

$$G^2 = ab,$$

so

$$G = \sqrt{4 \times 25} = 10.$$

The defining relation is

$$\frac{G}{4} = \frac{25}{G}$$

which gives

$$G^2 = 100.$$

Therefore,

$$\boxed{G = 10}.$$

SMART TIP

For a geometric mean G between a and b ,

$$\boxed{a, G, b}$$

must form a G.P.

Therefore,

$$\boxed{G^2 = ab}$$

is the safest way to remember the concept.

26 Insertion of Geometric Means

Suppose we want to insert m geometric means between a and b .

Then the complete G.P. is

$$a, ar, ar^2, \dots, ar^m, b.$$

There are $m + 2$ terms.

Therefore,

$$b = ar^{m+1}.$$

Hence,

$$r^{m+1} = \frac{b}{a}.$$

Therefore,

IMPORTANT FORMULA

$$\boxed{r = \left(\frac{b}{a}\right)^{\frac{1}{m+1}}}$$

26.1 Example 7

Insert two geometric means between 2 and 54.

SOLUTION

We need to insert 2 G.M.s.

Thus,

$$m = 2.$$

We have

$$a = 2, \quad b = 54.$$

Using

$$r = \left(\frac{b}{a}\right)^{1/(m+1)},$$

$$r = \left(\frac{54}{2}\right)^{1/3}$$

$$r = 27^{1/3}$$

$$r = 3.$$

Therefore, the G.P. is

$$2, 6, 18, 54.$$

Hence, the two geometric means are

$$\boxed{6, 18}.$$

26.2 Example 8

Insert three geometric means between 3 and 48.

SOLUTION

Here,

$$a = 3, \quad b = 48, \quad m = 3.$$

Therefore,

$$\begin{aligned} r &= \left(\frac{48}{3}\right)^{1/4} \\ &= 16^{1/4} \\ &= 2. \end{aligned}$$

Thus,

$$3, 6, 12, 24, 48$$

is the required G.P.

Therefore, the three geometric means are

$$\boxed{6, 12, 24}.$$

27 NCERT-Based Questions

27.1 Question 1

Determine whether

$$5, 15, 45, 135, \dots$$

is a G.P.

SOLUTION

We calculate the ratios:

$$\frac{15}{5} = 3,$$

$$\frac{45}{15} = 3,$$

$$\frac{135}{45} = 3.$$

All ratios are equal.

Therefore,

$$\boxed{\text{The sequence is a G.P.}}$$

with

$$\boxed{r = 3}.$$

27.2 Question 2

Find the 7th term of the G.P.

4, 12, 36, ...

SOLUTION

Here,

$$a = 4$$

and

$$r = \frac{12}{4} = 3.$$

Therefore,

$$a_7 = 4(3)^6.$$

Since

$$3^6 = 729,$$

we get

$$a_7 = 4(729)$$

Hence,

$$\boxed{a_7 = 2916}.$$

27.3 Question 3

Find the 9th term of the G.P.

128, 64, 32, ...

SOLUTION

Here,

$$a = 128$$

and

$$r = \frac{64}{128} = \frac{1}{2}.$$

Therefore,

$$a_9 = 128 \left(\frac{1}{2}\right)^8.$$

Since

$$128 = 2^7,$$

we get

$$a_9 = 2^7 \cdot 2^{-8}$$

$$= 2^{-1}.$$

Hence,

$$a_9 = \frac{1}{2}.$$

27.4 Question 4

Find the geometric mean between 12 and 75.

SOLUTION

Using

$$G = \sqrt{ab},$$

we get

$$G = \sqrt{12 \times 75}.$$

$$G = \sqrt{900}$$

Therefore,

$$G = 30.$$

27.5 Question 5

Insert three geometric means between 2 and 162.

SOLUTION

Here,

$$a = 2, \quad b = 162, \quad m = 3.$$

Therefore,

$$\begin{aligned} r &= \left(\frac{162}{2} \right)^{1/4} \\ &= 81^{1/4} \\ &= 3. \end{aligned}$$

Hence,

$$2, 6, 18, 54, 162.$$

Therefore, the three geometric means are

$$\boxed{6, 18, 54}.$$

28 A.P. vs G.P.

Feature	A.P.	G.P.
Pattern	Addition	Multiplication
Constant	Difference d	Ratio r
General term	$a + (n - 1)d$	ar^{n-1}
Example	2, 5, 8, 11	2, 6, 18, 54

SMART TIP

One-line memory trick:

$$\boxed{\text{A.P.} \Rightarrow \text{same difference}}$$

$$\boxed{\text{G.P.} \Rightarrow \text{same ratio}}$$

29 G.P. Formula Sheet

IMPORTANT FORMULA

First term:

$$a$$

Common ratio:

$$r = \frac{a_2}{a_1}$$

n th term:

$$a_n = ar^{n-1}$$

Geometric Mean:

$$G = \sqrt{ab}$$

m G.M.s between a and b :

$$r = \left(\frac{b}{a}\right)^{1/(m+1)}$$

Three numbers in G.P.:

$$\frac{a}{r}, a, ar$$

CHALLENGE QUESTION

The 5th term of a G.P. is 48 and the 8th term is 384. Find the common ratio.

Hint:

Use

$$a_n = ar^{n-1}$$

for both terms and divide the two equations.

NEXT PART

Sum of n Terms of a G.P.

Finite Geometric Series

Important NCERT Problems

30 Sum of n Terms of a G.P.

Consider the G.P.

$$a, ar, ar^2, ar^3, \dots$$

The sum of its first n terms is denoted by S_n .

Therefore,

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1}.$$

CONCEPT

For a G.P., we add terms that are obtained by repeatedly multiplying the previous term by the same number r .

For example,

2, 6, 18, 54, ...

has

$$a = 2, \quad r = 3.$$

30.1 Derivation of the Formula

Let

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1}.$$

Multiply both sides by r :

$$rS_n = ar + ar^2 + ar^3 + \cdots + ar^n.$$

Now subtract the first equation from the second:

$$rS_n - S_n = ar + ar^2 + \cdots + ar^n - (a + ar + ar^2 + \cdots + ar^{n-1}).$$

All middle terms cancel.

Therefore,

$$(r - 1)S_n = ar^n - a.$$

Hence,

$$S_n = \frac{ar^n - a}{r - 1}.$$

Taking a common,

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

for $r > 1$.

This can also be written as

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

for $r < 1$.

IMPORTANT FORMULA

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

or equivalently,

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

where

$$r \neq 1.$$

30.2 Which Form Should I Use?

Both formulas are exactly the same.

If $r > 1$, the form

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

is usually convenient.

If $0 < r < 1$, the form

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

is usually easier.

SMART TIP

Do not memorize two different formulas.

They are the **same formula** written in two forms.

$$\frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r}$$

31 Special Case: $r = 1$

If

$$r = 1,$$

then the G.P. becomes

$$a, a, a, a, \dots$$

Therefore,

$$S_n = a + a + \dots + a$$

with n terms.

Hence,

IMPORTANT FORMULA

When

$$r = 1,$$

we have

$$S_n = na.$$

31.1 Example 1

Find the sum of the first 8 terms of the G.P.

2, 6, 18, 54, ...

SOLUTION

Here,

$$a = 2, \quad r = 3, \quad n = 8.$$

Using

$$S_n = \frac{a(r^n - 1)}{r - 1},$$

we get

$$S_8 = \frac{2(3^8 - 1)}{3 - 1}.$$

Since

$$3^8 = 6561,$$

we get

$$\begin{aligned} S_8 &= \frac{2(6561 - 1)}{2} \\ &= 6560. \end{aligned}$$

Therefore,

$$S_8 = 6560.$$

31.2 Example 2

Find the sum

$1 + 2 + 4 + 8 + \cdots + 2^{10}.$

SOLUTION

The given series is

$$1 + 2 + 4 + 8 + \cdots + 2^{10}.$$

This is a G.P. with

$$a = 1, \quad r = 2.$$

The powers run from 2^0 to 2^{10} .

Therefore, the number of terms is

$$n = 11.$$

Using

$$S_n = \frac{a(r^n - 1)}{r - 1},$$

we get

$$\begin{aligned} S_{11} &= \frac{1(2^{11} - 1)}{2 - 1} \\ &= 2048 - 1. \end{aligned}$$

Hence,

$$\boxed{S_{11} = 2047}.$$

31.3 Example 3

Find the sum of the first 6 terms of

$$81 + 27 + 9 + 3 + \cdots.$$

SOLUTION

Here,

$$a = 81$$

and

$$r = \frac{27}{81} = \frac{1}{3}.$$

Also,

$$n = 6.$$

Using

$$S_n = \frac{a(1 - r^n)}{1 - r},$$

we get

$$S_6 = \frac{81 \left[1 - \left(\frac{1}{3} \right)^6 \right]}{1 - \frac{1}{3}}.$$

Since

$$1 - \frac{1}{3} = \frac{2}{3},$$

we get

$$\begin{aligned} S_6 &= 81 \left[1 - \frac{1}{729} \right] \frac{3}{2} \\ &= \frac{243}{2} \left(\frac{728}{729} \right). \end{aligned}$$

Therefore,

$$\boxed{S_6 = \frac{364}{3}}$$

32 G.P. with Negative Common Ratio

Consider

$$2, -4, 8, -16, \dots$$

Here,

$$a = 2, \quad r = -2.$$

The formula

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

works even when r is negative.

CONCEPT

There is no restriction that the common ratio must be positive.
The formula for the sum of a finite G.P. is valid for

$$\boxed{r \neq 1}.$$

Thus, r may be positive or negative.

32.1 Example 4

Find the sum of the first 5 terms of
 $2, -4, 8, -16, \dots$

SOLUTION

Here,

$$a = 2, \quad r = -2, \quad n = 5.$$

Using

$$\begin{aligned} S_n &= \frac{a(r^n - 1)}{r - 1}, \\ S_5 &= \frac{2[(-2)^5 - 1]}{-2 - 1} \\ &= \frac{2(-32 - 1)}{-3} \\ &= \frac{-66}{-3}. \end{aligned}$$

Therefore,

$$\boxed{S_5 = 22}.$$

Check directly:

$$2 - 4 + 8 - 16 + 32 = 22.$$

33 Sum When the Last Term is Given

Suppose the first term is a , last term is l , and common ratio is r .

We know

$$l = ar^{n-1}.$$

Therefore,

$$ar^n = lr.$$

Using

$$S_n = \frac{ar^n - a}{r - 1},$$

we get

$$S_n = \frac{lr - a}{r - 1}.$$

Hence,

IMPORTANT FORMULA

If the first term a , last term l and common ratio r are known,

$$S_n = \frac{lr - a}{r - 1}$$

for $r \neq 1$.

33.1 Example 5

Find the sum of the G.P.

$$3 + 6 + 12 + \cdots + 384.$$

SOLUTION

Here,

$$a = 3, \quad r = 2, \quad l = 384.$$

We can first find the number of terms.

$$l = ar^{n-1}$$

$$384 = 3(2^{n-1})$$

$$128 = 2^{n-1}$$

$$2^7 = 2^{n-1}.$$

Therefore,

$$n = 8.$$

Now,

$$S_n = \frac{n}{2}$$

cannot be used because that is an A.P. formula.

For a G.P.,

$$S_n = \frac{a(r^n - 1)}{r - 1}.$$

Hence,

$$S_8 = \frac{3(2^8 - 1)}{2 - 1}$$

$$= 3(256 - 1)$$

$$= 3(255).$$

Therefore,

$$\boxed{S_8 = 765}.$$

34 Finite Geometric Series

A finite G.P. has a fixed number of terms.

For example,

$$2 + 6 + 18 + 54$$

has four terms.

Its sum is

$$2 + 6 + 18 + 54 = 80.$$

Therefore,

$$S_4 = 80.$$

CONCEPT

Remember:

$$\text{G.P.} \rightarrow \text{sequence}$$

$$\text{Geometric Series} \rightarrow \text{sum of G.P. terms}$$

For example,

$$2, 6, 18, 54$$

is a G.P., whereas

$$2 + 6 + 18 + 54$$

is a geometric series.

35 NCERT-Based Questions

35.1 Question 1

Find the sum of the first 10 terms of the G.P.

$$1, 3, 9, 27, \dots$$

SOLUTION

Here,

$$a = 1, \quad r = 3, \quad n = 10.$$

Therefore,

$$S_{10} = \frac{1(3^{10} - 1)}{3 - 1}.$$

Since

$$3^{10} = 59049,$$

we get

$$\begin{aligned} S_{10} &= \frac{59049 - 1}{2} \\ &= \frac{59048}{2}. \end{aligned}$$

Hence,

$$S_{10} = 29524.$$

35.2 Question 2

Find the sum

$$5 + 10 + 20 + \cdots + 640.$$

SOLUTION

Here,

$$a = 5, \quad r = 2, \quad l = 640.$$

First find n :

$$640 = 5(2^{n-1})$$

$$128 = 2^{n-1}$$

$$2^7 = 2^{n-1}.$$

Therefore,

$$n = 8.$$

Now,

$$S_8 = \frac{5(2^8 - 1)}{2 - 1}.$$

$$= 5(256 - 1)$$

$$= 5(255).$$

Therefore,

$$\boxed{S_8 = 1275}.$$

35.3 Question 3

Find the sum of the first 7 terms of

$$64 + 32 + 16 + 8 + \cdots.$$

SOLUTION

Here,

$$a = 64, \quad r = \frac{1}{2}, \quad n = 7.$$

Using

$$S_n = \frac{a(1 - r^n)}{1 - r},$$

we get

$$\begin{aligned} S_7 &= \frac{64 \left[1 - \left(\frac{1}{2} \right)^7 \right]}{1 - \frac{1}{2}} \\ &= \frac{64 \left[1 - \frac{1}{128} \right]}{\frac{1}{2}} \\ &= 128 \left(\frac{127}{128} \right). \end{aligned}$$

Therefore,

$$\boxed{S_7 = 127}.$$

35.4 Question 4

Find the sum of the first 6 terms of the G.P.

$$3, -6, 12, -24, \dots$$

SOLUTION

Here,

$$a = 3$$

and

$$r = -2.$$

Also,

$$n = 6.$$

Using

$$S_n = \frac{a(r^n - 1)}{r - 1},$$

$$S_6 = \frac{3[(-2)^6 - 1]}{-2 - 1}.$$

$$= \frac{3(64 - 1)}{-3}$$

$$= \frac{189}{-3}.$$

Therefore,

$$\boxed{S_6 = -63}.$$

35.5 Question 5

The first term of a G.P. is 4 and its common ratio is 3. Find the sum of the first 5 terms.

SOLUTION

Given,

$$a = 4, \quad r = 3, \quad n = 5.$$

Using

$$S_n = \frac{a(r^n - 1)}{r - 1},$$

$$S_5 = \frac{4(3^5 - 1)}{3 - 1}.$$

Since

$$3^5 = 243,$$

we get

$$S_5 = \frac{4(243 - 1)}{2}$$

$$= 2(242)$$

Therefore,

$$\boxed{S_5 = 484}.$$

36 A.P. vs G.P. : Sum Formulas

Quantity	A.P.	G.P.
n th term	$a + (n - 1)d$	ar^{n-1}
Sum	$\frac{n}{2}[2a + (n - 1)d]$	$\frac{a(r^n - 1)}{r - 1}$
Constant	Difference d	Ratio r

SMART TIP**Golden Rule**

A.P. terms **difference**

G.P. terms **ratio**

$$\boxed{\text{A.P.} \Rightarrow \text{Add/Subtract}}$$

$$\boxed{\text{G.P.} \Rightarrow \text{Multiply/Divide}}$$

37 Useful Geometric Series

Some very common finite G.P. sums are:

IMPORTANT FORMULA

$$1 + r + r^2 + \cdots + r^{n-1} = \frac{r^n - 1}{r - 1}$$

for $r \neq 1$.

For $r = 2$,

$$1 + 2 + 2^2 + \cdots + 2^{n-1} = 2^n - 1.$$

Thus,

IMPORTANT FORMULA

$$1 + 2 + 4 + \cdots + 2^{n-1} = 2^n - 1$$

For $r = 3$,

$$1 + 3 + 9 + \cdots + 3^{n-1} = \frac{3^n - 1}{2}.$$

38 Common Mistakes

EXAM ALERT

1. Do not use the A.P. sum formula for a G.P.
2. In a G.P., identify r carefully.
3. Remember that

$$r \neq 1$$

for the formula

$$S_n = \frac{a(r^n - 1)}{r - 1}.$$

4. If $r = 1$, use

$$S_n = na.$$

5. Be careful with negative ratios. The sign of r^n depends on whether n is even or odd.

39 G.P. Master Formula Sheet

IMPORTANT FORMULA

First term

$$a$$

Common ratio

$$r = \frac{a_2}{a_1}$$

n th term

$$a_n = ar^{n-1}$$

Sum of first n terms

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

or

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

When $r = 1$

$$S_n = na$$

Geometric Mean

$$G = \sqrt{ab}$$

CHALLENGE QUESTION

The sum of the first 5 terms of a G.P. is 242 and its first term is 2. If the common ratio is positive, find r .

Hint:

Use

$$S_n = \frac{a(r^n - 1)}{r - 1}.$$

Put

$$S_5 = 242, \quad a = 2.$$

Then solve the resulting equation.

NEXT PART

Relationship Between A.M. and G.M.

$$\text{A.M.} \geq \text{G.M.}$$

Properties and NCERT Problems

40 Relationship Between A.M. and G.M.

We have already studied:

Arithmetic Mean $\rightarrow$ A.M.

and

Geometric Mean $\rightarrow$ G.M.

Now we will study the important relationship between them.

40.1 Arithmetic Mean of Two Positive Numbers

Let a and b be two positive numbers.

Their arithmetic mean is

IMPORTANT FORMULA

$$A = \frac{a+b}{2}$$

40.2 Geometric Mean of Two Positive Numbers

The geometric mean of a and b is

IMPORTANT FORMULA

$$G = \sqrt{ab}$$

Therefore,

$$A = \frac{a+b}{2}$$

and

$$G = \sqrt{ab}.$$

41 The Important Result

For any two positive real numbers a and b ,

IMPORTANT FORMULA

$$\frac{a+b}{2} \geq \sqrt{ab}$$

That is,

$$A \geq G$$

This is called the

A.M.–G.M. Inequality

41.1 Proof of A.M. $\geq$ G.M.

We know that the square of every real number is non-negative.

Therefore,

$$(a - b)^2 \geq 0.$$

Expanding,

$$a^2 - 2ab + b^2 \geq 0.$$

Hence,

$$a^2 + b^2 \geq 2ab.$$

For positive a and b , we can use

$$(a + b)^2 \geq 4ab.$$

Taking positive square roots,

$$a + b \geq 2\sqrt{ab}.$$

Dividing both sides by 2,

$$\frac{a + b}{2} \geq \sqrt{ab}.$$

Therefore,

IMPORTANT FORMULA

$$A \geq G$$

Hence proved.

42 When Does A.M. = G.M.?

We have

$$(a - b)^2 \geq 0.$$

Equality occurs when

$$(a - b)^2 = 0.$$

Therefore,

$$a - b = 0.$$

Hence,

$$a = b.$$

Thus,

IMPORTANT FORMULA

$$A = G \iff a = b$$

In other words, A.M. and G.M. are equal only when the two numbers are equal.

42.1 Example 1

Find the A.M. and G.M. of 4 and 16 and verify that A.M. $\geq$ G.M.

SOLUTION

Arithmetic Mean:

$$A = \frac{4 + 16}{2}$$

$$A = \frac{20}{2} = 10.$$

Therefore,

$$\boxed{A = 10}.$$

Geometric Mean:

$$G = \sqrt{4 \times 16}$$

$$= \sqrt{64} = 8.$$

Therefore,

$$\boxed{G = 8}.$$

Hence,

$$10 > 8.$$

Therefore,

$$\boxed{A > G}.$$

The A.M.–G.M. inequality is verified.

42.2 Example 2

Find the A.M. and G.M. of 9 and 9.

SOLUTION

A.M.:

$$A = \frac{9+9}{2} = 9.$$

G.M.:

$$G = \sqrt{9 \times 9} = 9.$$

Therefore,

$$\boxed{A = G = 9}.$$

This also verifies the equality condition.

Since the two numbers are equal,

$$\boxed{A = G}.$$

43 A Beautiful Way to Understand $A.M. \geq G.M.$

Suppose two positive numbers are a and b .

Their A.M. is

$$\frac{a+b}{2}.$$

Their G.M. is

$$\sqrt{ab}.$$

The inequality says

$$\boxed{\text{A.M. is always greater than or equal to G.M.}}$$

Think of it as:

Arithmetic Mean = average based on addition

Geometric Mean = average based on multiplication

For unequal positive numbers,

$$\boxed{A > G}.$$

For equal positive numbers,

$$\boxed{A = G}.$$

44 Application of $A.M. \geq G.M.$

The A.M.–G.M. inequality is extremely useful for finding maximum and minimum values.

Consider two positive numbers whose sum is fixed.

Suppose

$$x + y = 20.$$

Then

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

Therefore,

$$10 \geq \sqrt{xy}.$$

Squaring both sides,

$$100 \geq xy.$$

Hence,

IMPORTANT FORMULA

If

$$x + y = 20,$$

then

$$\boxed{xy \leq 100}.$$

The maximum value of xy is

$$\boxed{100}.$$

Equality occurs when

$$x = y = 10.$$

44.1 Example 3

If

$$x + y = 12,$$

find the maximum value of xy .

SOLUTION

By A.M.–G.M. inequality,

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

Since

$$x+y=12,$$

we get

$$\frac{12}{2} \geq \sqrt{xy}.$$

Therefore,

$$6 \geq \sqrt{xy}.$$

Squaring,

$$36 \geq xy.$$

Hence,

$$\boxed{xy \leq 36}.$$

Therefore, the maximum value of xy is

$$\boxed{36}.$$

Equality occurs when

$$x=y=6.$$

44.2 Example 4

If x and y are positive numbers such that

$$xy=25,$$

find the minimum value of $x+y$.

SOLUTION

Using A.M.–G.M.,

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

Since

$$xy = 25,$$

we get

$$\frac{x+y}{2} \geq 5.$$

Therefore,

$$x+y \geq 10.$$

Hence, the minimum value is

$$\boxed{10}.$$

Equality occurs when

$$x = y = 5.$$

SMART TIP**Very Important Pattern**

If the **sum** is fixed:

$$\boxed{x + y = \text{constant}}$$

then A.M.–G.M. can be used to find the **maximum product**.

If the **product** is fixed:

$$\boxed{xy = \text{constant}}$$

then A.M.–G.M. can be used to find the **minimum sum**.

45 A.M. $\geq$ G.M. for Three Positive Numbers

For three positive numbers a, b, c ,

IMPORTANT FORMULA

$$\boxed{\frac{a+b+c}{3} \geq \sqrt[3]{abc}}$$

Thus,

$$\boxed{\text{A.M.} \geq \text{G.M.}}$$

for three positive numbers also.

Equality occurs when

$$\boxed{a = b = c}.$$

45.1 Example 5

If x, y, z are positive numbers and

$$x + y + z = 15,$$

find the maximum value of xyz .

SOLUTION

By A.M.–G.M.,

$$\frac{x + y + z}{3} \geq \sqrt[3]{xyz}.$$

Since

$$x + y + z = 15,$$

we get

$$\frac{15}{3} \geq \sqrt[3]{xyz}.$$

Thus,

$$5 \geq \sqrt[3]{xyz}.$$

Cubing both sides,

$$125 \geq xyz.$$

Therefore,

$$\boxed{xyz \leq 125}.$$

Hence, the maximum value is

$$\boxed{125}.$$

Equality occurs when

$$x = y = z = 5.$$

46 A.M. $\geq$ G.M. for Four Positive Numbers

For four positive numbers a, b, c, d ,

IMPORTANT FORMULA

$$\frac{a + b + c + d}{4} \geq \sqrt[4]{abcd}$$

Equality occurs when

$$\boxed{a = b = c = d}.$$

47 NCERT-Based Questions

47.1 Question 1

Find the A.M. and G.M. of 8 and 18.

SOLUTION

A.M.:

$$\begin{aligned} A &= \frac{8 + 18}{2} \\ &= \frac{26}{2} = 13. \end{aligned}$$

G.M.:

$$\begin{aligned} G &= \sqrt{8 \times 18} \\ &= \sqrt{144} = 12. \end{aligned}$$

Therefore,

$$A = 13, \quad G = 12.$$

Hence,

$$A > G.$$

47.2 Question 2

If $x + y = 18$, find the maximum value of xy .

SOLUTION

Using

$$\frac{x+y}{2} \geq \sqrt{xy},$$

we get

$$\frac{18}{2} \geq \sqrt{xy}.$$

Thus,

$$9 \geq \sqrt{xy}.$$

Squaring,

$$81 \geq xy.$$

Therefore,

$$\boxed{\max(xy) = 81}.$$

Equality occurs at

$$\boxed{x = y = 9}.$$

47.3 Question 3

If $xy = 49$, find the minimum value of $x + y$ for positive x and y .

SOLUTION

By A.M.–G.M.,

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

Since

$$xy = 49,$$

we get

$$\frac{x+y}{2} \geq 7.$$

Therefore,

$$x + y \geq 14.$$

Hence,

$$\boxed{\min(x+y) = 14}.$$

Equality occurs when

$$\boxed{x = y = 7}.$$

47.4 Question 4

If a, b, c are positive numbers such that

$$a + b + c = 12,$$

find the maximum value of abc .

SOLUTION

Using

$$\frac{a + b + c}{3} \geq \sqrt[3]{abc},$$

we get

$$\frac{12}{3} \geq \sqrt[3]{abc}.$$

Therefore,

$$4 \geq \sqrt[3]{abc}.$$

Cubing,

$$64 \geq abc.$$

Hence,

$$\boxed{\max(abc) = 64}.$$

Equality occurs when

$$\boxed{a = b = c = 4}.$$

47.5 Question 5

Prove that for positive numbers a and b ,

$$a + b \geq 2\sqrt{ab}.$$

SOLUTION

We know that

$$(a - b)^2 \geq 0.$$

Therefore,

$$a^2 - 2ab + b^2 \geq 0.$$

Hence,

$$(a + b)^2 \geq 4ab.$$

Since a and b are positive,

$$a + b \geq 2\sqrt{ab}.$$

Therefore,

$$\boxed{a + b \geq 2\sqrt{ab}}$$

Hence proved.

48 Equality Condition

The equality condition is one of the most important things to remember.

IMPORTANT FORMULA

For positive numbers,

$$\boxed{A = G \iff \text{all the numbers are equal}}$$

For two numbers:

$$\boxed{a = b}$$

For three numbers:

$$\boxed{a = b = c}$$

For four numbers:

$$\boxed{a = b = c = d}$$

49 A.M.–G.M. Master Sheet

IMPORTANT FORMULA

For two positive numbers a, b :

$$A = \frac{a+b}{2}$$

$$G = \sqrt{ab}$$

and

$$A \geq G$$

Therefore,

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Equality occurs when

$$a = b.$$

For three positive numbers:

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

Equality occurs when

$$a = b = c.$$

EXAM TRICK

Question: $x + y$ is fixed. What should you think?

Maximum product

Question: xy is fixed. What should you think?

Minimum sum

Question: Equality ?

$$x = y$$

CHALLENGE QUESTION

If x, y, z are positive numbers such that

$$x + y + z = 18,$$

find the maximum value of

$$xyz.$$

Also state when equality occurs.

NEXT PART

Special Series
 $\sum k, \sum k^2, \sum k^3$
 and NCERT Exercise Problems

50 Special Series

In the previous sections, we studied A.P. and G.P.

Now we will study some very important series whose sums can be written directly using standard formulas.

These are called **special series**.

50.1 Sum of First n Natural Numbers

Consider the series

$$1 + 2 + 3 + 4 + \cdots + n.$$

This is an A.P. with

$$a = 1, \quad d = 1.$$

Using the formula for the sum of an A.P.,

$$S_n = \frac{n}{2}[2a + (n-1)d].$$

Therefore,

$$\begin{aligned} S_n &= \frac{n}{2}[2 + (n-1)] \\ &= \frac{n}{2}(n+1). \end{aligned}$$

Hence,

IMPORTANT FORMULA

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

50.2 Example 1

Find

$$1 + 2 + 3 + \cdots + 100.$$

SOLUTION

Using

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2},$$

put $n = 100$.

Therefore,

$$\begin{aligned} S_{100} &= \frac{100(101)}{2} \\ &= 50(101). \end{aligned}$$

Hence,

$$\boxed{S_{100} = 5050}.$$

51 Sum of First n Even Numbers

The first n even numbers are

$$2, 4, 6, 8, \dots, 2n.$$

Therefore,

$$2 + 4 + 6 + \cdots + 2n$$

is an A.P.

We can write

$$2 + 4 + 6 + \cdots + 2n = 2(1 + 2 + 3 + \cdots + n).$$

Using

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2},$$

we obtain

$$2 + 4 + \cdots + 2n = 2 \left[\frac{n(n+1)}{2} \right].$$

Therefore,

IMPORTANT FORMULA

$$\boxed{2 + 4 + 6 + \cdots + 2n = n(n+1)}$$

51.1 Example 2

Find the sum of the first 50 even numbers.

SOLUTION

Using

$$2 + 4 + \cdots + 2n = n(n + 1),$$

put

$$n = 50.$$

Therefore,

$$S = 50(51).$$

Hence,

$$\boxed{S = 2550}.$$

52 Sum of First n Odd Numbers

The first n odd numbers are

$$1, 3, 5, \dots, (2n - 1).$$

This is an A.P. with

$$a = 1, \quad d = 2.$$

Using

$$S_n = \frac{n}{2}[2a + (n - 1)d],$$

we get

$$S_n = \frac{n}{2}[2 + 2(n - 1)].$$

Therefore,

$$\begin{aligned} S_n &= \frac{n}{2}(2n) \\ &= n^2. \end{aligned}$$

Hence,

IMPORTANT FORMULA

$$\boxed{1 + 3 + 5 + \cdots + (2n - 1) = n^2}$$

CONCEPT

A beautiful pattern:

$$1 = 1^2$$

$$1 + 3 = 4 = 2^2$$

$$1 + 3 + 5 = 9 = 3^2$$

$$1 + 3 + 5 + 7 = 16 = 4^2$$

Therefore,

$$\boxed{\text{Sum of first } n \text{ odd numbers} = n^2}$$

52.1 Example 3

Find the sum of the first 25 odd numbers.

SOLUTION

Using

$$1 + 3 + \cdots + (2n - 1) = n^2,$$

we put

$$n = 25.$$

Therefore,

$$S = 25^2.$$

Hence,

$$\boxed{S = 625}.$$

53 Sum of Squares of First n Natural Numbers

Consider

$$1^2 + 2^2 + 3^2 + \cdots + n^2.$$

This series has the important formula

IMPORTANT FORMULA

$$\boxed{1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}}$$

This formula should be remembered carefully.

53.1 Example 4

Find

$$1^2 + 2^2 + 3^2 + \cdots + 10^2.$$

SOLUTION

Using

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

put

$$n = 10.$$

Therefore,

$$S = \frac{10(11)(21)}{6}.$$

Simplifying,

$$S = \frac{5(11)(21)}{3}$$

$$= 5(11)(7)$$

$$= 385.$$

Hence,

$$\boxed{S = 385}.$$

54 Sum of Cubes of First n Natural Numbers

Consider

$$1^3 + 2^3 + 3^3 + \cdots + n^3.$$

The beautiful formula is

IMPORTANT FORMULA

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

In other words,

$$\boxed{1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2}$$

This is a very beautiful result.

54.1 Example 5

Find

$$1^3 + 2^3 + 3^3 + 4^3 + 5^3.$$

SOLUTION

Here,

$$n = 5.$$

Using

$$1^3 + 2^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2} \right]^2,$$

we get

$$\begin{aligned} S &= \left[\frac{5(6)}{2} \right]^2 \\ &= 15^2. \end{aligned}$$

Therefore,

$$\boxed{S = 225}.$$

Check:

$$1 + 8 + 27 + 64 + 125 = 225.$$

CONCEPT

Notice the beautiful relationship:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Therefore,

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2.$$

So,

$$\boxed{\sum k^3 = \left(\sum k \right)^2}$$

55 Sigma Notation

In mathematics, instead of writing a long sum, we often use the symbol

$$\boxed{\Sigma}$$

which is called the **sigma notation**.

For example,

$$1 + 2 + 3 + \cdots + n$$

can be written as

$$\sum_{k=1}^n k$$

Similarly,

$$1^2 + 2^2 + \cdots + n^2$$

can be written as

$$\sum_{k=1}^n k^2$$

and

$$1^3 + 2^3 + \cdots + n^3$$

can be written as

$$\sum_{k=1}^n k^3$$

56 Standard Summation Formulas

IMPORTANT FORMULA

$$\sum_{k=1}^n 1 = n$$

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

56.1 Example 6

Evaluate

$$\sum_{k=1}^{20} k.$$

SOLUTION

Using

$$\sum_{k=1}^n k = \frac{n(n+1)}{2},$$

we get

$$\sum_{k=1}^{20} k = \frac{20(21)}{2}.$$

Therefore,

$$\boxed{210}.$$

56.2 Example 7

Evaluate

$$\sum_{k=1}^{10} k^2.$$

SOLUTION

Using

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6},$$

we get

$$\sum_{k=1}^{10} k^2 = \frac{10(11)(21)}{6}.$$

Therefore,

$$\boxed{385}.$$

56.3 Example 8

Evaluate

$$\sum_{k=1}^8 k^3.$$

SOLUTION

Using

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2,$$

we get

$$\begin{aligned} \sum_{k=1}^8 k^3 &= \left[\frac{8(9)}{2} \right]^2 \\ &= 36^2. \end{aligned}$$

Hence,

$$\boxed{1296}.$$

57 Sums Involving a Constant

Suppose we have

$$\sum_{k=1}^n c$$

where c is a constant.

Since c occurs n times,

IMPORTANT FORMULA

$$\sum_{k=1}^n c = nc$$

For example,

$$\sum_{k=1}^{10} 5 = 5 + 5 + \cdots + 5$$

with 10 terms.

Therefore,

$$\boxed{50}.$$

58 Sum of a Linear Expression

Consider

$$\sum_{k=1}^n (ak + b).$$

Using the distributive property,

$$\sum_{k=1}^n (ak + b) = a \sum_{k=1}^n k + b \sum_{k=1}^n 1.$$

Therefore,

IMPORTANT FORMULA

$$\sum_{k=1}^n (ak + b) = a \frac{n(n+1)}{2} + bn$$

58.1 Example 9

Evaluate

$$\sum_{k=1}^{10} (2k + 3).$$

SOLUTION

We write

$$\sum_{k=1}^{10} (2k + 3) = 2 \sum_{k=1}^{10} k + 3 \sum_{k=1}^{10} 1.$$

Now,

$$\sum_{k=1}^{10} k = \frac{10(11)}{2} = 55.$$

Also,

$$\sum_{k=1}^{10} 1 = 10.$$

Therefore,

$$\begin{aligned} & 2(55) + 3(10) \\ &= 110 + 30 \\ &= 140. \end{aligned}$$

Hence,

$$\boxed{140}.$$

58.2 Example 10

Evaluate

$$\sum_{k=1}^{20} (3k - 2).$$

SOLUTION

Using

$$\sum_{k=1}^n (ak + b) = a \frac{n(n+1)}{2} + bn,$$

we have

$$a = 3, \quad b = -2, \quad n = 20.$$

Therefore,

$$S = 3 \frac{20(21)}{2} - 2(20).$$

$$= 3(210) - 40.$$

$$= 630 - 40.$$

Hence,

$$\boxed{S = 590}.$$

59 NCERT-Based Questions

59.1 Question 1

Find

$$1 + 2 + 3 + \cdots + 50.$$

SOLUTION

Using

$$\frac{n(n+1)}{2},$$

we get

$$S = \frac{50(51)}{2}.$$

Therefore,

$$\boxed{S = 1275}.$$

59.2 Question 2

Find

$$2 + 4 + 6 + \cdots + 100.$$

SOLUTION

The last even number is

$$100 = 2n.$$

Therefore,

$$n = 50.$$

Using

$$2 + 4 + \cdots + 2n = n(n + 1),$$

we get

$$S = 50(51).$$

Hence,

$$\boxed{S = 2550}.$$

59.3 Question 3

Find

$$1 + 3 + 5 + \cdots + 99.$$

SOLUTION

The last odd number is

$$99 = 2n - 1.$$

Therefore,

$$2n = 100$$

and

$$n = 50.$$

Using

$$1 + 3 + \cdots + (2n - 1) = n^2,$$

we get

$$S = 50^2.$$

Hence,

$$\boxed{S = 2500}.$$

59.4 Question 4

Find

$$1^2 + 2^2 + \cdots + 20^2.$$

SOLUTION

Using

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6},$$

we get

$$S = \frac{20(21)(41)}{6}.$$

Simplifying,

$$\begin{aligned} S &= \frac{10(21)(41)}{3} \\ &= 10(7)(41). \end{aligned}$$

Therefore,

$$\boxed{S = 2870}.$$

59.5 Question 5

Find

$$1^3 + 2^3 + \cdots + 10^3.$$

SOLUTION

Using

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2,$$

we get

$$\begin{aligned} S &= \left[\frac{10(11)}{2} \right]^2 \\ &= 55^2. \end{aligned}$$

Therefore,

$$\boxed{S = 3025}.$$

59.6 Question 6

Evaluate

$$\sum_{k=1}^{15} (4k + 1).$$

SOLUTION

We have

$$\sum_{k=1}^{15} (4k + 1) = 4 \sum_{k=1}^{15} k + \sum_{k=1}^{15} 1.$$

Now,

$$\sum_{k=1}^{15} k = \frac{15(16)}{2} = 120.$$

Therefore,

$$S = 4(120) + 15.$$

$$= 480 + 15.$$

Hence,

$$\boxed{S = 495}.$$

60 Special Series Formula Table

Series	Sum
$1 + 2 + \cdots + n$	$\frac{n(n+1)}{2}$
$2 + 4 + \cdots + 2n$	$n(n+1)$
$1 + 3 + \cdots + (2n-1)$	n^2
$1^2 + 2^2 + \cdots + n^2$	$\frac{n(n+1)(2n+1)}{6}$
$1^3 + 2^3 + \cdots + n^3$	$\left[\frac{n(n+1)}{2} \right]^2$

MEMORY TRICK

Remember the first three important formulas:

$$\boxed{\sum k = \frac{n(n+1)}{2}}$$

$$\boxed{\sum k^2 = \frac{n(n+1)(2n+1)}{6}}$$

$$\boxed{\sum k^3 = \left[\frac{n(n+1)}{2} \right]^2}$$

Notice:

$$\boxed{\sum k^3 = (\sum k)^2}$$

CHALLENGE QUESTION

Evaluate

$$1 + 2 + 3 + \cdots + 100$$

without adding the terms one by one.

Then find

$$1^3 + 2^3 + 3^3 + \cdots + 100^3.$$

61 Special Series – Quick Revision**IMPORTANT FORMULA**

$$\sum_{k=1}^n 1 = n$$

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

NEXT PART

Complete Chapter Revision

Important NCERT Exercise Problems

Mixed Questions + Conceptual Problems

62 Mixed NCERT Practice**Sequence and Series – Concept Builder**

“Question concept , formula .”

62.1 Question 1

Find the 20th term of the sequence

$$3, 7, 11, 15, \dots$$

SOLUTION

The difference between consecutive terms is

$$7 - 3 = 4,$$

$$11 - 7 = 4.$$

Therefore, it is an A.P.

Hence,

$$a = 3, \quad d = 4.$$

The n th term of an A.P. is

$$a_n = a + (n - 1)d.$$

Therefore,

$$a_{20} = 3 + (20 - 1)(4).$$

$$= 3 + 76.$$

Hence,

$$\boxed{a_{20} = 79}.$$

62.2 Question 2

Find the 15th term of the G.P.

$$2, 6, 18, 54, \dots$$

SOLUTION

Here,

$$a = 2$$

and

$$r = \frac{6}{2} = 3.$$

For a G.P.,

$$a_n = ar^{n-1}.$$

Therefore,

$$a_{15} = 2(3)^{14}.$$

Hence,

$$\boxed{a_{15} = 2 \cdot 3^{14}}.$$

62.3 Question 3

Which term of the A.P.

$$7, 12, 17, 22, \dots$$

is 72?

SOLUTION

Here,

$$a = 7, \quad d = 5.$$

We need

$$a_n = 72.$$

Using

$$a_n = a + (n - 1)d,$$

we get

$$72 = 7 + (n - 1)5.$$

Therefore,

$$65 = 5(n - 1).$$

Thus,

$$n - 1 = 13.$$

Hence,

$$\boxed{n = 14}.$$

Therefore, 72 is the

$$\boxed{14\text{th term}}.$$

62.4 Question 4

Which term of the G.P.

$$3, 6, 12, 24, \dots$$

is 384?

SOLUTION

Here,

$$a = 3, \quad r = 2.$$

We need

$$3(2^{n-1}) = 384.$$

Therefore,

$$2^{n-1} = 128.$$

Since

$$128 = 2^7,$$

we get

$$n - 1 = 7.$$

Hence,

$$\boxed{n = 8}.$$

Therefore, 384 is the

$$\boxed{8\text{th term}}.$$

62.5 Question 5

Find the sum of the first 20 terms of

$$5, 8, 11, 14, \dots$$

SOLUTION

This is an A.P. because the common difference is

$$d = 3.$$

Thus,

$$a = 5, \quad d = 3, \quad n = 20.$$

Using

$$S_n = \frac{n}{2}[2a + (n - 1)d],$$

we get

$$S_{20} = \frac{20}{2}[2(5) + 19(3)].$$

$$= 10[10 + 57].$$

$$= 10(67).$$

Hence,

$$\boxed{S_{20} = 670}.$$

62.6 Question 6

Find the sum of the first 8 terms of

$$2, 6, 18, 54, \dots$$

SOLUTION

This is a G.P. with

$$a = 2, \quad r = 3, \quad n = 8.$$

Using

$$S_n = \frac{a(r^n - 1)}{r - 1},$$

we get

$$S_8 = \frac{2(3^8 - 1)}{3 - 1}.$$

Since

$$3^8 = 6561,$$

we get

$$S_8 = \frac{2(6560)}{2}.$$

Therefore,

$$\boxed{S_8 = 6560}.$$

62.7 Question 7

Find the arithmetic mean between 18 and 42.

SOLUTION

The arithmetic mean of two numbers a and b is

$$A = \frac{a + b}{2}.$$

Therefore,

$$\begin{aligned} A &= \frac{18 + 42}{2} \\ &= \frac{60}{2}. \end{aligned}$$

Hence,

$$\boxed{A = 30}.$$

62.8 Question 8

Find the geometric mean between 4 and 25.

SOLUTION

The geometric mean is

$$G = \sqrt{ab}.$$

Therefore,

$$G = \sqrt{4 \times 25}$$

$$= \sqrt{100}.$$

Hence,

$$\boxed{G = 10}.$$

62.9 Question 9

Insert three arithmetic means between 5 and 25.

SOLUTION

We need the A.P.

$$5, A_1, A_2, A_3, 25.$$

There are 5 terms.

Therefore,

$$a = 5, \quad a_5 = 25.$$

Using

$$a_n = a + (n - 1)d,$$

we get

$$25 = 5 + 4d.$$

Thus,

$$20 = 4d.$$

Therefore,

$$d = 5.$$

Hence the A.P. is

$$5, 10, 15, 20, 25.$$

Therefore, the three arithmetic means are

$$\boxed{10, 15, 20}.$$

62.10 Question 10

Insert two geometric means between 4 and 108.

SOLUTION

We need

$$4, G_1, G_2, 108.$$

This is a G.P. with four terms.
Therefore,

$$a = 4, \quad a_4 = 108.$$

Using

$$a_n = ar^{n-1},$$

we get

$$108 = 4r^3.$$

Thus,

$$r^3 = 27.$$

Hence,

$$r = 3.$$

Therefore, the G.P. is

$$4, 12, 36, 108.$$

Hence, the required geometric means are

$$\boxed{12, 36}.$$

62.11 Question 11

If the 5th term of an A.P. is 18 and the 9th term is 30, find its first term and common difference.

SOLUTION

For an A.P.,

$$a_n = a + (n - 1)d.$$

For the 5th term,

$$a + 4d = 18.$$

For the 9th term,

$$a + 8d = 30.$$

Subtracting,

$$4d = 12.$$

Therefore,

$$d = 3.$$

Substituting in

$$a + 4d = 18,$$

we get

$$a + 12 = 18.$$

Thus,

$$a = 6.$$

Hence,

IMPORTANT FORMULA

$$a = 6, \quad d = 3$$

62.12 Question 12

If the 3rd term of a G.P. is 12 and the 5th term is 48, find the common ratio.

SOLUTION

For a G.P.,

$$a_n = ar^{n-1}.$$

Therefore,

$$a_3 = ar^2 = 12$$

and

$$a_5 = ar^4 = 48.$$

Dividing,

$$\frac{ar^4}{ar^2} = \frac{48}{12}.$$

Hence,

$$r^2 = 4.$$

Therefore,

$$\boxed{r = \pm 2}.$$

If the question assumes a positive common ratio, then

$$\boxed{r = 2}.$$

62.13 Question 13

Find the sum

$$1 + 2 + 3 + \cdots + 50.$$

SOLUTION

Using

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2},$$

we get

$$S = \frac{50(51)}{2}.$$

Therefore,

$$\boxed{S = 1275}.$$

62.14 Question 14

Find

$$1^2 + 2^2 + 3^2 + \cdots + 20^2.$$

SOLUTION

Using

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6},$$

we get

$$S = \frac{20(21)(41)}{6}.$$

Therefore,

$$\boxed{S = 2870}.$$

62.15 Question 15

Find

$$1^3 + 2^3 + 3^3 + \cdots + 15^3.$$

SOLUTION

Using

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2,$$

we get

$$\begin{aligned} S &= \left[\frac{15(16)}{2} \right]^2 \\ &= 120^2. \end{aligned}$$

Therefore,

$$\boxed{S = 14400}.$$

62.16 Question 16

If $x, y > 0$ and

$$x + y = 20,$$

find the maximum value of xy .

SOLUTION

By A.M.–G.M.,

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

Therefore,

$$10 \geq \sqrt{xy}.$$

Squaring,

$$100 \geq xy.$$

Hence,

$$\boxed{\max(xy) = 100}.$$

Equality occurs when

$$\boxed{x = y = 10}.$$

62.17 Question 17

If $x, y > 0$ and

$$xy = 64,$$

find the minimum value of $x + y$.

SOLUTION

Using A.M.–G.M.,

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

Therefore,

$$\frac{x+y}{2} \geq 8.$$

Thus,

$$x + y \geq 16.$$

Hence,

$$\boxed{\min(x+y) = 16}.$$

Equality occurs when

$$\boxed{x = y = 8}.$$

62.18 Question 18

Determine whether the sequence

$$1, 4, 9, 16, 25, \dots$$

is an A.P. or a G.P.

SOLUTION

Check the differences:

$$4 - 1 = 3,$$

$$9 - 4 = 5,$$

$$16 - 9 = 7.$$

The differences are not equal.

Therefore, it is not an A.P.

Now check ratios:

$$\frac{4}{1} \neq \frac{9}{4}.$$

Therefore, it is not a G.P. either.

Hence,

IMPORTANT FORMULA

The sequence is neither A.P. nor G.P.

62.19 Question 19

If the 7th term of an A.P. is 20 and the 13th term is 38, find the 20th term.

SOLUTION

We have

$$a_7 = a + 6d = 20$$

and

$$a_{13} = a + 12d = 38.$$

Subtracting,

$$6d = 18.$$

Therefore,

$$d = 3.$$

From

$$a + 6d = 20,$$

we get

$$a + 18 = 20.$$

Thus,

$$a = 2.$$

Now,

$$a_{20} = a + 19d.$$

Therefore,

$$a_{20} = 2 + 19(3)$$

$$= 2 + 57.$$

Hence,

$$\boxed{a_{20} = 59}.$$

62.20 Question 20

Find the sum of the first 10 terms of the A.P.

$$7, 10, 13, 16, \dots$$

SOLUTION

Here,

$$a = 7, \quad d = 3, \quad n = 10.$$

Using

$$S_n = \frac{n}{2}[2a + (n - 1)d],$$

we get

$$S_{10} = \frac{10}{2}[14 + 9(3)].$$

$$= 5[14 + 27].$$

$$= 5(41).$$

Therefore,

$$\boxed{S_{10} = 205}.$$

63 How to Identify the Question

QUESTION IDENTIFICATION MAP

Step 1: Look at the sequence.

If

$$a_2 - a_1 = a_3 - a_2 = \dots$$

then

A.P.

If

$$\frac{a_2}{a_1} = \frac{a_3}{a_2} = \dots$$

then

G.P.

Step 2: What is being asked?

Term $\rightarrow a_n$

Sum $\rightarrow S_n$

Means $\rightarrow A.M.$ or $G.M.$

Maximum/Minimum $\rightarrow A.M. \geq G.M.$

64 Complete Formula Wall

IMPORTANT FORMULA

A.P.

$$a_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$S_n = \frac{n}{2}(a + l)$$

G.P.

$$a_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

A.M. and G.M.

$$A = \frac{a + b}{2}$$

$$G = \sqrt{ab}$$

$$A \geq G$$

Special Series

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

65 DAV sector-4 students Practice Set

TRY YOURSELF

1. Find the 25th term of the A.P.

$$4, 9, 14, 19, \dots$$

2. Find the sum of the first 15 terms of

$$3, 7, 11, 15, \dots$$

3. Find the 10th term of the G.P.

$$5, 10, 20, 40, \dots$$

4. Find the sum of the first 7 terms of

$$1, 2, 4, 8, \dots$$

5. Insert four arithmetic means between 6 and 36.

6. Insert three geometric means between 2 and 162.

7. Find

$$1 + 2 + \dots + 75.$$

8. Find

$$1^2 + 2^2 + \dots + 15^2.$$

9. Find

$$1^3 + 2^3 + \dots + 12^3.$$

10. If $x + y = 30$, find the maximum value of xy .

SMART STUDY TIP

Sequence and Series formulas .
question :

A.P. G.P.?

:

Term Sum?

:

- formula ?

-step approach questions automatically .

NEXT PART

NCERT Exercise – Question by Question Solution
Concept → Formula → Solution

66 Important NCERT Questions**SEQUENCE AND SERIES**

Important NCERT-Based Questions with Short Solutions

Question 1

Find the 20th term of the A.P.

7, 10, 13, 16, ...

SOLUTION

Here,

$$a = 7, \quad d = 3.$$

Using

$$a_n = a + (n - 1)d,$$

we get

$$a_{20} = 7 + 19(3)$$

$$= 7 + 57.$$

Therefore,

$$a_{20} = 64.$$

Question 2

Which term of the A.P.

$$3, 8, 13, 18, \dots$$

is 78?

SOLUTION

Here,

$$a = 3, \quad d = 5.$$

Let the required term be the n th term.

Then

$$78 = 3 + (n - 1)5.$$

Therefore,

$$75 = 5(n - 1).$$

Hence,

$$n - 1 = 15.$$

Thus,

$$\boxed{n = 16}.$$

Therefore, 78 is the

$$\boxed{16\text{th term}}.$$

Question 3

Find the sum of the first 25 terms of

$$2, 5, 8, 11, \dots$$

SOLUTION

Here,

$$a = 2, \quad d = 3, \quad n = 25.$$

Using

$$S_n = \frac{n}{2}[2a + (n - 1)d],$$

we get

$$S_{25} = \frac{25}{2}[4 + 24(3)].$$

$$= \frac{25}{2}(76).$$

Therefore,

$$\boxed{S_{25} = 950}.$$

Question 4

Find the sum of all natural numbers between 100 and 200 which are divisible by 5.

SOLUTION

The required numbers are

$$105, 110, 115, \dots, 195.$$

This is an A.P. with

$$a = 105, \quad d = 5, \quad l = 195.$$

First find the number of terms:

$$n = \frac{l - a}{d} + 1.$$

Therefore,

$$\begin{aligned} n &= \frac{195 - 105}{5} + 1 \\ &= 18 + 1 = 19. \end{aligned}$$

Now,

$$S_n = \frac{n}{2}(a + l).$$

Thus,

$$\begin{aligned} S &= \frac{19}{2}(105 + 195) \\ &= \frac{19}{2}(300). \end{aligned}$$

Hence,

$$\boxed{S = 2850}.$$

Question 5

Find the 15th term of the G.P.

$$2, 6, 18, 54, \dots$$

SOLUTION

Here,

$$a = 2, \quad r = 3.$$

For a G.P.,

$$a_n = ar^{n-1}.$$

Therefore,

$$a_{15} = 2(3)^{14}.$$

Hence,

$$\boxed{a_{15} = 2 \cdot 3^{14}}.$$

Question 6

Find the sum of the first 8 terms of the G.P.

$$3, 6, 12, 24, \dots$$

SOLUTION

Here,

$$a = 3, \quad r = 2, \quad n = 8.$$

Using

$$S_n = \frac{a(r^n - 1)}{r - 1},$$

we get

$$S_8 = \frac{3(2^8 - 1)}{2 - 1}.$$

$$= 3(256 - 1).$$

$$= 3(255).$$

Therefore,

$$\boxed{S_8 = 765}.$$

Question 7

Insert five arithmetic means between 8 and 38.

SOLUTION

We need the A.P.

$$8, A_1, A_2, A_3, A_4, A_5, 38.$$

There are 7 terms.

Thus,

$$a = 8, \quad a_7 = 38.$$

Using

$$a_n = a + (n - 1)d,$$

we get

$$38 = 8 + 6d.$$

Hence,

$$30 = 6d$$

and

$$d = 5.$$

Therefore, the A.P. is

$$8, 13, 18, 23, 28, 33, 38.$$

Hence, the required means are

$$\boxed{13, 18, 23, 28, 33}.$$

Question 8

Insert three geometric means between 2 and 162.

SOLUTION

We need

$$2, G_1, G_2, G_3, 162.$$

There are 5 terms.

For a G.P.,

$$a_5 = ar^4.$$

Therefore,

$$162 = 2r^4.$$

Hence,

$$r^4 = 81.$$

Taking the positive ratio,

$$r = 3.$$

Therefore, the G.P. is

$$2, 6, 18, 54, 162.$$

Hence,

$$\boxed{G_1 = 6, \quad G_2 = 18, \quad G_3 = 54}.$$

Question 9

The 7th term of an A.P. is 20 and the 13th term is 38. Find the first term and common difference.

SOLUTION

We have

$$a_7 = a + 6d = 20$$

and

$$a_{13} = a + 12d = 38.$$

Subtracting,

$$6d = 18.$$

Thus,

$$d = 3.$$

Now,

$$a + 6(3) = 20.$$

Therefore,

$$a = 2.$$

Hence,

$$\boxed{a = 2, \quad d = 3}.$$

Question 10

The 3rd term of a G.P. is 12 and the 6th term is 96. Find the common ratio.

SOLUTION

For a G.P.,

$$a_n = ar^{n-1}.$$

Therefore,

$$ar^2 = 12$$

and

$$ar^5 = 96.$$

Dividing,

$$\frac{ar^5}{ar^2} = \frac{96}{12}.$$

Thus,

$$r^3 = 8.$$

Hence,

$$\boxed{r = 2}.$$

Question 11

Find the sum

$$1 + 2 + 3 + \cdots + 100.$$

SOLUTION

Using

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2},$$

we get

$$S = \frac{100(101)}{2}.$$

Therefore,

$$\boxed{S = 5050}.$$

Question 12

Find

$$1^2 + 2^2 + 3^2 + \cdots + 20^2.$$

SOLUTION

Using

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6},$$

we get

$$S = \frac{20(21)(41)}{6}.$$

Therefore,

$$\boxed{S = 2870}.$$

Question 13

Find

$$1^3 + 2^3 + 3^3 + \cdots + 10^3.$$

SOLUTION

Using

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2,$$

we get

$$\begin{aligned} S &= \left[\frac{10(11)}{2} \right]^2 \\ &= 55^2. \end{aligned}$$

Hence,

$$\boxed{S = 3025}.$$

Question 14

If $x + y = 20$, find the maximum value of xy .

SOLUTION

By A.M.–G.M.,

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

Therefore,

$$10 \geq \sqrt{xy}.$$

Squaring,

$$100 \geq xy.$$

Hence,

$$\boxed{\max(xy) = 100}.$$

Equality occurs when

$$\boxed{x = y = 10}.$$

Question 15

If $xy = 81$, find the minimum value of $x + y$ for positive x and y .

SOLUTION

By A.M.–G.M.,

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

Therefore,

$$\frac{x+y}{2} \geq 9.$$

Thus,

$$x + y \geq 18.$$

Hence,

$$\boxed{\min(x+y) = 18}.$$

Equality occurs when

$$\boxed{x = y = 9}.$$

Question 16

Find the sum of the first 20 odd natural numbers.

SOLUTION

The first n odd numbers have sum

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

For $n = 20$,

$$S = 20^2.$$

Therefore,

$$\boxed{S = 400}.$$

Question 17

Find the sum of the first 20 even natural numbers.

SOLUTION

We know

$$2 + 4 + 6 + \cdots + 2n = n(n + 1).$$

For $n = 20$,

$$S = 20(21).$$

Therefore,

$$\boxed{S = 420}.$$

Question 18

If the n th term of a sequence is

$$a_n = 3n - 5,$$

find its 10th term.

SOLUTION

Put

$$n = 10.$$

Then,

$$a_{10} = 3(10) - 5.$$

$$= 30 - 5.$$

Hence,

$$\boxed{a_{10} = 25}.$$

Question 19

Show that the sequence

$$5, 9, 13, 17, \dots$$

is an A.P. Find its 25th term.

SOLUTION

We have

$$9 - 5 = 4,$$

$$13 - 9 = 4,$$

$$17 - 13 = 4.$$

Since the common difference is constant,

The sequence is an A.P.

Here,

$$a = 5, \quad d = 4.$$

Therefore,

$$a_{25} = 5 + 24(4).$$

$$= 5 + 96.$$

Hence,

$a_{25} = 101$.

Question 20

Find the sum of the first 10 terms of the A.P.

$$5, 9, 13, 17, \dots$$

SOLUTION

Here,

$$a = 5, \quad d = 4, \quad n = 10.$$

Using

$$S_n = \frac{n}{2}[2a + (n-1)d],$$

we get

$$S_{10} = \frac{10}{2}[10 + 9(4)].$$

$$= 5(46).$$

Therefore,

$$\boxed{S_{10} = 230}.$$

67 Most Important Questions for Exam

questions practice :

MUST DO QUESTIONS

1. Find the n th term of an A.P.
2. Find which term of an A.P. is equal to a given number.
3. Find the sum of the first n terms of an A.P.
4. Find the sum of all multiples of a number within a given interval.
5. Find the n th term of a G.P.
6. Find the sum of the first n terms of a G.P.
7. Insert arithmetic means between two given numbers.
8. Insert geometric means between two given numbers.
9. Find a and d when two terms of an A.P. are given.
10. Find r when two terms of a G.P. are given.
11. Use

$$\frac{a+b}{2} \geq \sqrt{ab}$$

to find maximum or minimum values.

12. Evaluate

$$\sum k, \quad \sum k^2, \quad \sum k^3.$$

68 Final Formula Revision

IMPORTANT FORMULA

A.P.

$$a_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$S_n = \frac{n}{2}(a + l)$$

G.P.

$$a_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

A.M.–G.M.

$$\frac{a + b}{2} \geq \sqrt{ab}$$

Special Series

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

GOLDEN TIP

Sequence and Series question :

$$\boxed{\text{A.P. G.P.}}$$

:

$$\boxed{\text{Term Sum}}$$

maximum/minimum :

$$\boxed{\text{A.M.} \geq \text{G.M.}}$$

questions